

Notes: Chetty, Friedman, Olsen, and Pistaferri (QJE, 2011)

Adjustment Costs, Firm Responses, and Micro vs. Macro Labor Supply Elasticities: Evidence from Danish Tax Records

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Administrative Danish tax data	3
3.2	Tax schedule and incentive variation	3
3.3	Main outcomes	3
3.4	Bunching and excess mass	3
3.5	Kink size and scope	4
4	Models and empirical specifications	4
4.1	Frictionless benchmark	4
4.2	Search costs and the inaction region	4
4.3	Micro versus macro elasticity	4
4.4	Firm and union responses	5
4.5	Tax reform regression	5
5	Identification and empirical design	5
5.1	Size of the tax change	5
5.2	Scope of affected workers	5
5.3	Occupation-level tests	5
5.4	Self-employed placebo group	6
5.5	Limitations	6
6	Section-by-section study guide	6
7	Tables and figures	7
7.1	Figure I: Bunching at Kinks with Search Costs + Figure II: The Danish Income Tax System	7
7.2	Figure III: Income Distributions around the Top Tax Cutoff for Wage Earners + Table I: Summary Statistics for the Danish Population and Estimation Samples, 1994–2001	7
7.3	Figure IV: Income Distributions around the Top Tax Cutoff, 1994–2001 + Figure V: Distinguishing Changes in Tax Incentives from Inflation and Wage Growth	8
7.4	Figure VI: Income Distributions around the Middle Tax Cutoff + Table II: Observed Elasticity Estimates Using Small Tax Reforms	9
7.5	Figure VII: Teachers’ Wage Earnings Distributions + Figure VIII: Modes of Occupation-Level Wage Earnings Distributions	10

7.6	Figure IX: Individual vs. Aggregate Bunching at the Pension Kink + Figure X: Observed Elasticities vs. Scope of Tax Changes	11
7.7	Figure XI: Dynamics of Earnings Around the Top Tax Cutoff + Figure XII: Correlation Between Individual and Aggregate Bunching	12
7.8	Figure XIII: Self-Employed Individuals	13
8	Detailed result map	14
9	Short summary	14
10	Conclusion	15
10.1	Core conclusion	15
10.2	Economic intuition	15
10.3	Contribution to public finance	15
10.4	Contribution to labor economics	15
10.5	Policy implication	15
10.6	Limitation	15
10.7	Takeaway	16

1 Big picture

- **Question.** Why do quasi-experimental micro estimates of intensive-margin labor supply elasticities often look small, while macro comparisons and bunching around large kinks suggest larger responses?
- **Core answer.** The paper argues that observed labor-supply and taxable-income responses depend on adjustment costs for workers and hours constraints or compensation packages set by firms and unions.
- **Main empirical setting.** The authors use Danish administrative tax records from 1994–2001, a period with a large top tax kink and smaller middle tax kinks.
- **Central mechanism.** When workers face search costs, small changes in incentives do not justify changing jobs or hours packages. Large kinks or broad policy changes can overcome frictions and also induce firms/unions to adjust job offers.
- **Micro versus macro distinction.** Micro estimates from small reforms or narrow kinks are attenuated because they measure ex-post fine-tuning around a current job. Macro elasticities can be larger because they reflect ex-ante job search and equilibrium firm responses.
- **Key evidence.** Wage earners bunch sharply at the large top tax kink, bunch little at the smaller middle kink, and respond little to small tax reforms. Self-employed individuals, who face fewer hours constraints, bunch strongly at more kinks.
- **Firm-response evidence.** Occupation-level wage modes cluster around the statutory top tax kink, and wage earners with deductions still bunch at the statutory kink even when their own personal tax kink is elsewhere.
- **Economic takeaway.** The same tax change can generate very different observed elasticities depending on the size of the tax change, the number of affected workers, and whether firms or institutions adjust labor-market options.

2 Incremental contribution of the paper

- **Reconciles micro and macro elasticity evidence.** The paper provides a concrete mechanism for why micro elasticities from tax reforms can be small even when steady-state or macro elasticities are materially larger.
- **Moves bunching analysis beyond frictionless models.** Standard bunching methods treat a kink as revealing a structural elasticity. This paper shows that bunching can be attenuated by optimization frictions and amplified by firm responses.
- **Develops size and scope predictions.** The paper does not merely claim frictions matter; it derives testable predictions: observed elasticities rise with tax-change size, rise with the scope of workers affected, and correlate with aggregate bunching across occupations.
- **Integrates worker frictions and firm-side constraints.** The model links individual adjustment costs with endogenous hours constraints or job packages posted by firms and unions.
- **Uses rich administrative data to test mechanisms.** Danish tax records allow the authors to compare large and small kinks, wage earners and self-employed workers, occupations, teachers, and pension-driven individual kinks.
- **Separates individual bunching from aggregate bunching.** The pension kink and occupation-level wage modes isolate individual job-search responses from firm/union supply-side responses.
- **Clarifies policy interpretation.** The paper warns that quasi-experiments affecting small groups may underestimate the long-run effects of broad tax reforms when institutions and firms adjust.

3 Data and measurement

3.1 Administrative Danish tax data

- The empirical analysis uses matched administrative tax records for the Danish population from 1994 to 2001.
- The data include taxable income, wage earnings, deductions, pension contributions, demographic characteristics, and occupation information.
- The setting is valuable because the Danish tax schedule has large, visible kinks in a central region of the income distribution.

Interpretation: The administrative data allow the authors to study bunching in the full population rather than a survey sample, making it possible to detect even modest excess mass around tax thresholds.

3.2 Tax schedule and incentive variation

- Denmark had a three-bracket income tax system during the sample period.
- The middle tax kink reduces the net-of-tax rate by roughly 11%.
- The top tax kink reduces the net-of-tax rate by roughly 30%.
- The top cutoff moves over time in nominal and real terms, which helps distinguish tax responses from inflation or wage-growth patterns.

Interpretation: The top kink is large enough to overcome frictions for some workers. The middle kink and small tax reforms provide contrast cases where standard incentives exist but adjustment costs dominate.

3.3 Main outcomes

- **Taxable income:** the primary concept for tax incentives and bunching.
- **Wage earnings:** used to separate real labor-market responses from pension shifting or other taxable-income manipulation.
- **Net deductions:** used to determine whether workers reach the top tax bracket at the statutory cutoff or at an individual-specific wage-earnings level.
- **Pension contributions:** used to construct the pension kink that helps separate individual from aggregate bunching.

Interpretation: Wage earnings are crucial because they are third-party reported and are less vulnerable to evasion than self-employment income.

3.4 Bunching and excess mass

- The authors estimate excess mass around kinks by comparing the observed distribution near the kink to a smooth counterfactual distribution.
- The counterfactual is fitted by excluding bins close to the kink and fitting a polynomial through the surrounding distribution.
- The excess mass is normalized relative to the counterfactual density near the kink.

Interpretation: Bunching is interpreted as the number of individuals who move to the kink because marginal tax rates change at that point.

3.5 Kink size and scope

- **Size** refers to the change in the marginal net-of-tax rate at the kink.
- **Scope** refers to the number or share of workers affected by a given kink or tax incentive.
- The paper compares the large top kink, the smaller middle kink, and small tax reforms.

Interpretation: Size tests whether incentives must be large enough to overcome individual search costs. Scope tests whether firms and unions adjust job offers when many workers share the same tax incentives.

4 Models and empirical specifications

4.1 Frictionless benchmark

In a frictionless model, workers choose hours continuously. A kink in the budget set generates bunching, and bunching can be mapped into a structural elasticity.

$$B \approx h_0(z^*)\Delta z^*, \quad \hat{e} \approx \frac{\Delta z^*/z^*}{\Delta \log(1 - \tau)}.$$

Interpretation: If the model were frictionless, small and large kinks would reveal the same underlying elasticity after scaling by the net-of-tax-rate change.

4.2 Search costs and the inaction region

Workers pay a fixed cost to find a different job or hours package. Around a kink, the worker only moves if the utility gain exceeds the search cost.

$$U(c, h) - \phi \geq U(c_0, h_0).$$

This creates an inaction band around the kink:

$$\underline{h} < h < \bar{h}.$$

Interpretation: Workers inside the inaction band do not relocate to the kink even if the kink changes their marginal incentive. Large tax changes shrink the inaction region and generate more bunching.

4.3 Micro versus macro elasticity

The paper distinguishes between elasticities identified from small policy changes and elasticities relevant for broad steady-state comparisons.

$$\hat{\varepsilon}^{MAC} = \frac{E \log h_i(\tau') - E \log h_i(\tau)}{\log(1 - \tau') - \log(1 - \tau)} \approx \varepsilon.$$

Interpretation: Macro variation changes the jobs workers search for and the job packages firms offer. Micro variation after a worker has already chosen a job is attenuated by search costs.

4.4 Firm and union responses

Firms and unions post or negotiate job offers. The equilibrium distribution of hours and compensation packages reflects the aggregate distribution of workers' preferences and tax incentives.

$$\text{job-offer distribution} = G(h; \text{aggregate tax incentives}).$$

Interpretation: If many workers face the same statutory kink, firms and unions may offer jobs that place workers near that kink. This generates aggregate bunching even for workers whose individual kink differs.

4.5 Tax reform regression

The small-reform evidence is estimated with specifications of the form:

$$\Delta \log z_{it} = \alpha + e \Delta \log(1 - \tau_{it}) + X'_{it} \gamma + \varepsilon_{it},$$

where the tax-rate change is instrumented using simulated tax rates based on base-year variables.

Interpretation: These regressions identify responses to relatively small tax changes. Near-zero estimates are consistent with adjustment costs attenuating standard micro estimates.

5 Identification and empirical design

5.1 Size of the tax change

- The top tax kink generates a large fall in the net-of-tax rate.
- The middle kink and small reforms generate smaller incentives.
- If frictionless heterogeneity alone were responsible, observed elasticities should not systematically rise with kink size after controlling for income and groups.

Interpretation: The large top kink provides the strongest test of whether sizable incentives can overcome adjustment frictions.

5.2 Scope of affected workers

- Most wage earners have small deductions and therefore face the statutory top tax cutoff in wage earnings.
- Workers with large deductions or non-wage income have individual kinks at other wage levels.
- If wages cluster at the statutory kink even for workers whose own kink is elsewhere, the clustering is likely to come from firm/union job packages.

Interpretation: This is the paper's clearest evidence of aggregate bunching: the supply side of the labor market adapts to common tax incentives.

5.3 Occupation-level tests

- Occupation-level wage modes are used to identify whether job offers cluster near tax thresholds.
- Teachers are especially useful because wages are collectively bargained and many teachers face common tax incentives.
- The pension kink helps measure individual bunching separately from aggregate bunching.

Interpretation: These tests link the model to institutional wage setting and show that bunching is not only an individual optimization response.

5.4 Self-employed placebo group

- Self-employed individuals can adjust reported income more easily and face fewer firm-imposed hours constraints.
- They bunch sharply at both large and small kinks.
- They do not show the same scope patterns as wage earners.

Interpretation: The self-employed results are a placebo for the friction mechanism: when frictions are low, size and scope matter less.

5.5 Limitations

- The paper does not structurally estimate the underlying macro elasticity.
- The bunching approach measures taxable-income responses, not hours directly.
- Some self-employed bunching may involve evasion or income shifting.
- Occupation-level correlations are not randomized causal estimates, but mechanism evidence.

Interpretation: The paper is strongest as a mechanism and reconciliation paper: it shows how frictions and firm responses shape observed elasticities.

6 Section-by-section study guide

- **Introduction.** Explains the micro versus macro elasticity puzzle and lays out the three predictions.
- **Model.** Builds a labor supply framework with search costs and endogenous hours constraints.
- **Institutional background and data.** Describes Danish taxes, the large top kink, the smaller middle kink, and administrative records.
- **Empirical analysis.** Tests the predictions using bunching, small tax reforms, occupation-level wage modes, the pension kink, and self-employed placebo tests.
- **Conclusion.** Interprets the findings as evidence that small quasi-experiments may understate steady-state responses.

7 Tables and figures

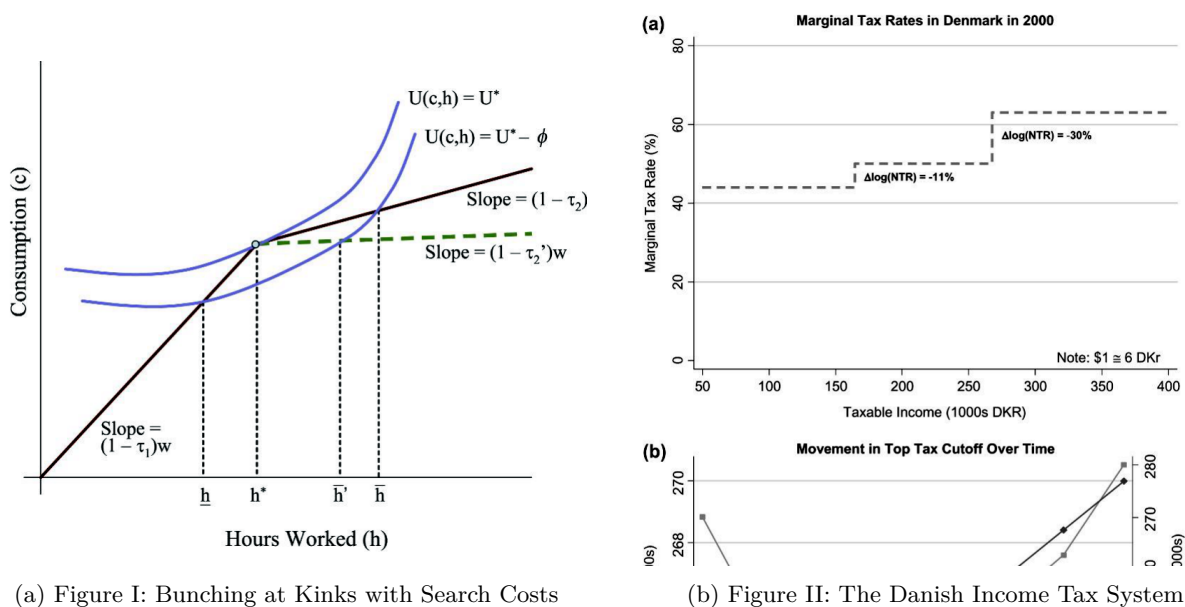
7.1 Figure I: Bunching at Kinks with Search Costs + Figure II: The Danish Income Tax System

Read:

- Figure I shows how a kink changes the budget constraint and how search costs create an inaction band.
- The worker who would optimally move to the kink may not move when the gain from adjustment is smaller than the search cost.
- A larger tax change steepens the budget-set kink and shrinks the inaction band.
- Figure II shows that the Danish top tax kink is much larger than the middle tax kink.
- The top cutoff also moves over time, which helps separate tax incentives from inflation or wage growth.

Detailed interpretation: Figure I is the conceptual foundation of the paper. It shows why bunching is not simply a sufficient statistic for the structural elasticity when workers face search costs. Figure II then shows why Denmark is a useful setting: the same tax system contains both small and large kinks.

Economic message: A kink must be large enough to overcome adjustment costs before bunching reveals meaningful labor-supply responsiveness.



7.2 Figure III: Income Distributions around the Top Tax Cutoff for Wage Earners + Table I: Summary Statistics for the Danish Population and Estimation Samples, 1994–2001

Read:

- Figure IIIa shows visually evident excess mass at the top tax cutoff in the full wage-earner sample.
- Married women and teachers show stronger bunching than single men and military workers.

- Table I shows that the estimation samples are large and rich enough to support fine distributional analysis.
- The wage-earner sample has millions of observations and detailed income, deduction, and tax-payment information.

Detailed interpretation: Figure III establishes the central empirical fact: wage earners bunch at the large top kink. The subgroup panels show that bunching is not uniform across groups, consistent with differences in frictions and institutional constraints. Table I clarifies the population and sample definitions behind the figures.

Economic message: Large administrative datasets make it possible to detect localized bunching and compare groups with different adjustment frictions.

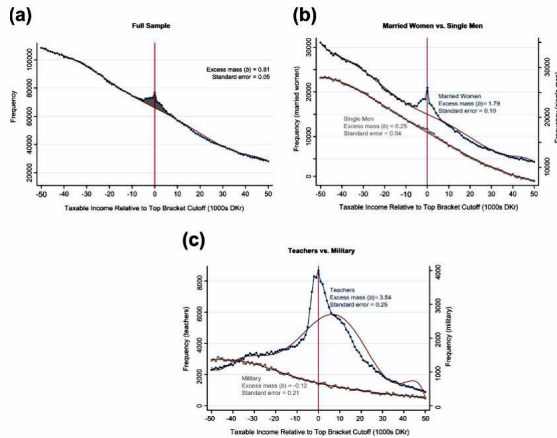


FIGURE III

Income Distributions around the Top Tax Cutoff for Wage Earners

TABLE I
SUMMARY STATISTICS FOR THE DANISH POPULATION AND ESTIMATION SAMPLES,
1994–2001

	Population (1)	Wage Earners (2)	Wage Earners < DKr 50,000 from top tax cutoff (3)	Self employed (4)
Demographics:				
Age	40.91	39.17	41.43	46.02
Children	0.62	0.67	0.71	0.70
Labor market experience (yrs)	12.46	15.42	18.77	9.46
College education (%)	17.61	22.76	28.54	17.74
Female (%)	49.61	48.17	39.17	24.40
Married (%)	50.62	53.64	58.68	67.34
Income:				
Wage earnings	149,254	236,478	269,340	38,343
Other personal income	42,642	9,408	2,747	153,467
Total personal income	180,213	227,359	251,145	188,854
Net capital income	-10,672	-15,819	-19,570	-7,785
Deductions:				
Net deductions	-40,687	-13,151	-6,381	-31,996
[Net deductions] < 7,500 (%)	43.25	59.36	69.11	23.84
[Net deductions-Pension kink] < 7,500 (%)	2.03	2.72	2.96	5.07
Individual pension contributions	4,316	4,217	4,535	16,709
Employer pension contributions	7,584	12,065	13,131	2,123
Tax Payments:				
Pred. liability accurate within 5 DKr (%)	95.11	94.83	94.47	93.62
Pays the middle tax (%)	50.38	74.23	95.57	45.48
Pays the top tax (%)	18.06	25.87	33.53	23.61
2-year growth in NTR (%)	1.68	2.25	2.25	1.07
Std dev of 2-year growth in NTR (%)	4.50	4.95	4.95	6.80
Number of Obs.	30,492,819	17,866,090	6,788,235	1,846,064

Notes. Table entries are means unless otherwise noted. Column 1 is based on the full population of Denmark between ages 15-70 from 1994-2001. Column 2 includes all wage earners, the primary estimation sample. Column 3 includes only the subset of wage earners for whom [taxable income - top tax cutoff] < 50,000, i.e. the individuals in Figure 3. Column 4 considers individuals who report positive self-employment income. All monetary values are in real 2000 Danish Kroner. Children are the number of children younger than 18 living with the individual. Personal income refers to all non-capital income. Net capital income refers to capital income minus payments such as mortgage interest. Net deductions refer to deductions from the top tax base such as individual pension contributions minus non-wage income such as taxable gifts. Net of tax rate is one minus the marginal tax rate predicted by our tax simulator.

(a) Figure III: Income Distributions around the Top Tax Cutoff for Wage Earners

(b) Table I: Summary Statistics

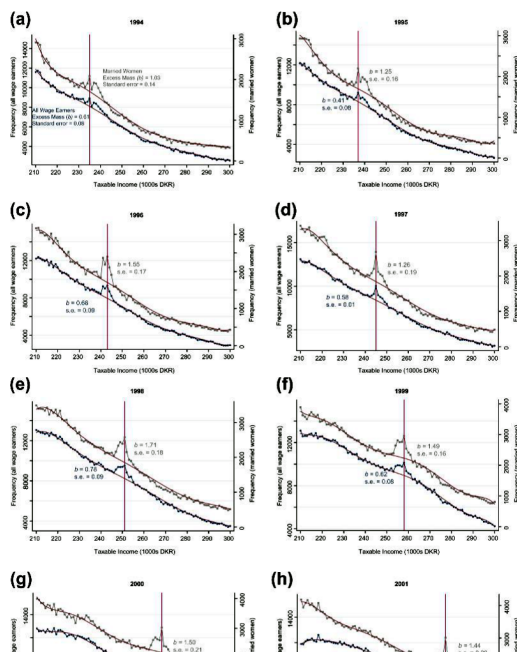
7.3 Figure IV: Income Distributions around the Top Tax Cutoff, 1994–2001 + Figure V: Distinguishing Changes in Tax Incentives from Inflation and Wage Growth

Read:

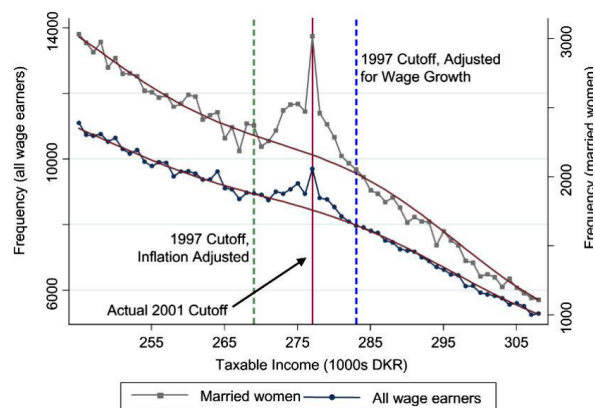
- Figure IV shows that bunching at the top tax kink is present across years.
- The location of excess mass moves with the nominal top tax cutoff.
- Figure V zooms in on 2001 and compares the actual cutoff with 1997 cutoffs adjusted for inflation and wage growth.
- The excess mass aligns with the tax cutoff, not with inflation-adjusted or wage-growth-adjusted historical thresholds.

Detailed interpretation: These figures address a crucial alternative interpretation: that bunching is driven by nominal wage norms or inflation rather than tax incentives. The movement of the mass with tax law supports the tax-incentive interpretation.

Economic message: The bunching responds to policy thresholds rather than to mechanical wage growth or nominal inertia.



(a) Figure IV: Top Tax Cutoff, 1994–2001



(b) Figure V: Tax Incentives versus Inflation and Wage Growth

7.4 Figure VI: Income Distributions around the Middle Tax Cutoff + Table II: Observed Elasticity Estimates Using Small Tax Reforms

Read:

- Figure VI shows little bunching at the smaller middle tax kink.
- The middle tax kink is a genuine tax incentive, but the change in the net-of-tax rate is much smaller than at the top kink.
- Table II shows that small reform estimates are close to zero across specifications and subgroups.
- The interaction with income in Table II does not support the idea that high-income workers are simply more elastic.

Detailed interpretation: Together these results support Prediction 1: observed elasticities rise with the size of tax variation. Small reforms and small kinks do not induce large observed responses even though the same individuals can respond to larger kinks.

Economic message: Standard micro estimates based on small incentives can be attenuated by frictions and may understate steady-state responses.

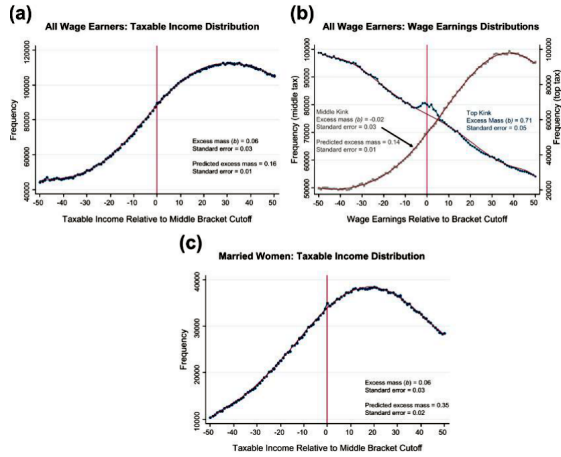


FIGURE VI

(a) Figure VI: Income Distributions around the Middle Tax Cutoff

TABLE II
OBSERVED ELASTICITY ESTIMATES USING SMALL TAX REFORMS

Subgroup:	Dependent Variable: Log Change in Wage Earnings					
	(1)	(2)	(3)	(4)	(5)	(6)
Log change in net-of-tax rate $\Delta \log(1-t)$	-0.001 (0.003)	-0.004 (0.003)	0.006 (0.005)	0.000 (0.011)	-0.006 (0.003)	-0.001 (0.003)
Earnings $> \Delta \log(1-t)$					0.005 (0.008)	
Labor income spline	x	x	x	x	x	x
Personal income spline	x	x	x	x	x	x
Year fixed effects	x	x	x	x	x	x
Age fixed effects	x	x	x	x	x	x
Region fixed effects					x	
Occupation fixed effects					x	
Capital income spline					x	
Gender, Marital status					x	
Observations	11,682,651	8,302,905	3,203,515	159,661	7,967,169	7,585,011

Note: Standard errors clustered by individual reported in parentheses. Dependent variable in all specifications is two-year growth rate in real wage earnings. Independent variable of interest is two-year growth rate in net-of-tax rate, instrumented using two-year growth rate in simulated net-of-tax rate using base-year variables. Coefficients reported can be interpreted as observed wage earnings elasticities from tax reforms. All specifications include 10-piece wage earnings and total personal income splines as well as age and year fixed effects. Column 1 also includes a 10-piece capital income spline, gender and marital status indicators, and region and occupation fixed effects. Occupation fixed effects are available only for a subset of years and observations. Column 4 restricts attention to married female professionals with more than 10 years of labor market experience. Column 5 interacts the log change in net-of-tax rate with the difference between wage earnings and the top tax cutoff (measured in DFR 100,000) to test whether the taxable income elasticity varies by income level. This specification restricts the sample to individuals with wage earnings in the base year between DFR 100,000 and 200,000. Column 6 considers individuals with more than DFR 200,000 of wage earnings in the base year.

(b) Table II: Observed Elasticity Estimates Using Small Tax Reforms

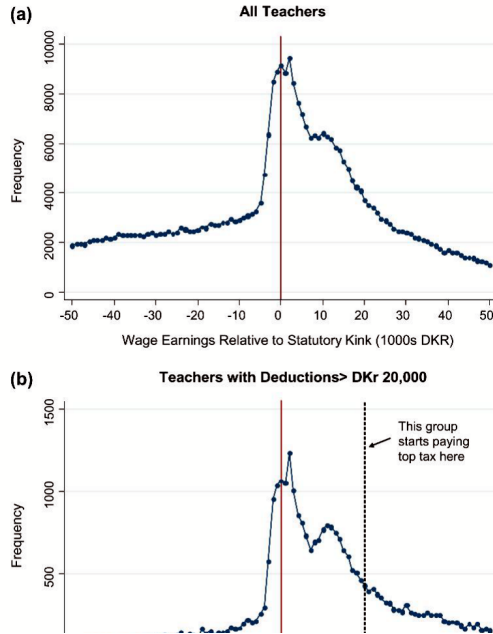
7.5 Figure VII: Teachers' Wage Earnings Distributions + Figure VIII: Modes of Occupation-Level Wage Earnings Distributions

Read:

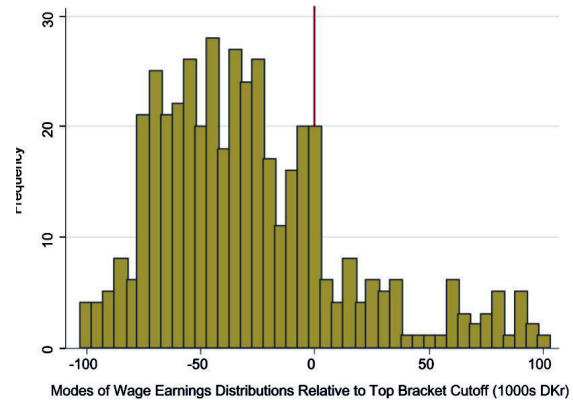
- Figure VII shows strong bunching among teachers around the statutory top tax kink.
- Teachers with large deductions also show the statutory kink pattern, even when their own individual tax kink differs.
- Figure VIII shows that occupation-level wage modes are disproportionately located below the statutory top tax cutoff.
- This pattern points to firm/union wage packages, not only individual worker choices.

Detailed interpretation: These figures provide the strongest evidence for aggregate bunching. When the wage distribution or occupation-level modes align with the common statutory kink, the supply side of the labor market appears to be responding to the tax preferences of the majority group.

Economic message: Firms and unions can amplify tax responses by tailoring job offers to common tax thresholds.



(a) Figure VII: Teachers' Wage Earnings Distributions



(b) Figure VIII: Occupation-Level Wage Modes

7.6 Figure IX: Individual vs. Aggregate Bunching at the Pension Kink + Figure X: Observed Elasticities vs. Scope of Tax Changes

Read:

- Figure IX uses the pension kink to separate individual bunching from aggregate bunching.
- Workers with large deductions bunch at the pension kink, showing individual search/adjustment responses.
- Workers without the relevant pension incentive do not bunch at that pension kink, supporting the interpretation.
- Figure X shows that observed elasticities rise with the fraction of the group that has small deductions, i.e., with the commonness of the tax incentive.

Detailed interpretation: Figure IX demonstrates that individual bunching is real but limited to workers with the relevant personal incentives. Figure X shows the scope effect: the more common a tax incentive is within a group, the stronger the observed elasticity. This is exactly what the endogenous hours-constraint model predicts.

Economic message: The tax response is larger when more workers share the same tax incentive because firms and unions have stronger incentives to adapt job offers.

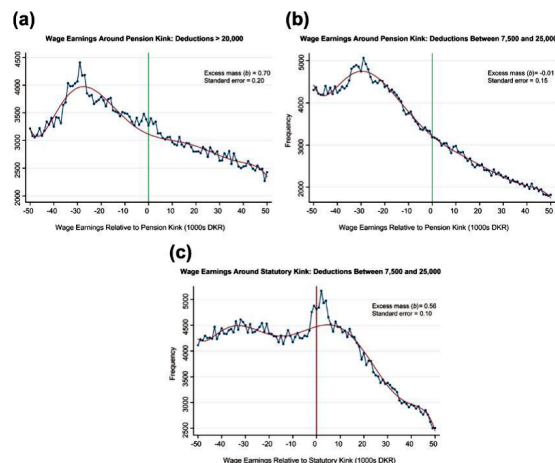
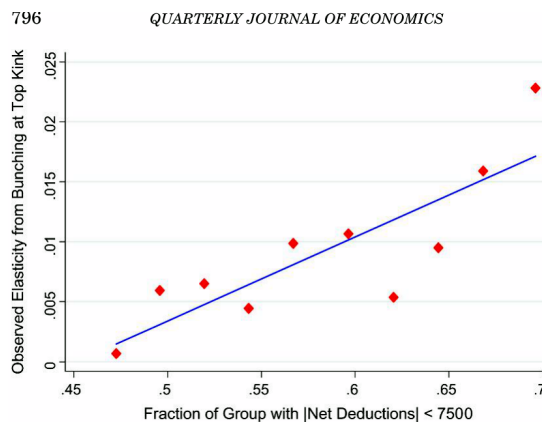


FIGURE IX
Individual vs. Aggregate Bunching at the Pension Kink
Panel (a) plots the distribution of wage earnings relative to the pension kink

(a) Figure IX: Individual vs. Aggregate Bunching at the Pension Kink



(b) Figure X: Observed Elasticities vs. Scope of Tax Changes

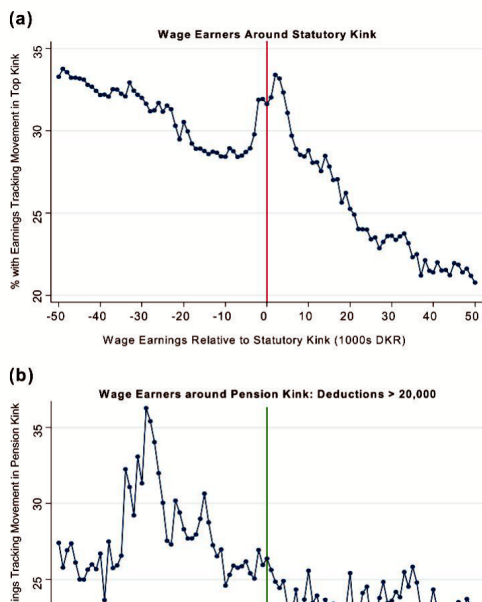
7.7 Figure XI: Dynamics of Earnings Around the Top Tax Cutoff + Figure XII: Correlation Between Individual and Aggregate Bunching

Read:

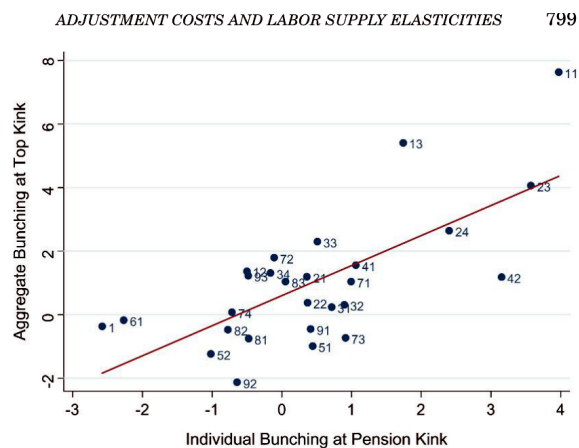
- Figure XI shows that earnings dynamics around the statutory top kink track the movement of the kink.
- For the pension kink, the dynamics are less consistent with aggregate firm adjustment.
- Figure XII shows that occupations with more individual bunching at the pension kink also have more aggregate bunching at the statutory top kink.
- The correlation supports Prediction 3, although it is not a randomized causal estimate.

Detailed interpretation: Figure XI helps distinguish individual adjustment from firm-level wage package adjustment over time. Figure XII links the two margins: occupations in which workers individually respond more also exhibit stronger aggregate wage-setting responses.

Economic message: Individual preferences and firm/union wage-setting institutions interact in equilibrium; observed bunching is partly a social or institutional response.



(a) Figure XI: Dynamics of Earnings Around the Top Tax Cutoff



(b) Figure XII: Individual and Aggregate Bunching

7.8 Figure XIII: Self-Employed Individuals

Read:

- The self-employed bunch sharply at the top tax cutoff.
- They also bunch at the middle tax cutoff, unlike wage earners.
- The self-employed with large deductions do not show aggregate bunching at the statutory wage-earner kink.
- The relationship between observed elasticities and scope is flat for the self-employed.

Detailed interpretation: The self-employed serve as a placebo group because they face fewer hours constraints and can adjust reported income more flexibly. Their behavior shows that taxpayers do respond when frictions are low, but the wage-earner patterns depend on labor-market constraints and firm responses.

Economic message: Frictions attenuate observed elasticities for wage earners; when frictions are lower, size and scope matter less.

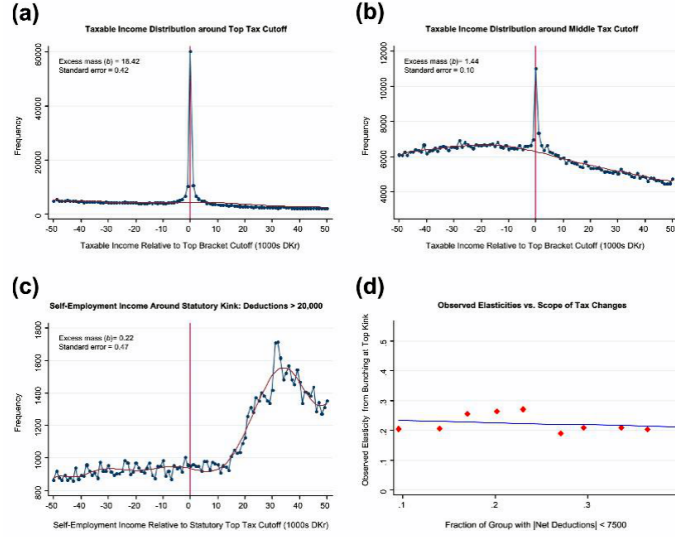


FIGURE XIII
 Self-Employed Individuals
 These figures include only individuals who report positive self-employment

Figure XIII: Self-Employed Individuals

8 Detailed result map

- **Result 1: Large kinks generate more bunching.** The top tax kink produces visible excess mass, while the smaller middle kink produces little bunching.
- **Result 2: Small reforms produce near-zero elasticities.** Table II shows that reform-based elasticities are close to zero, reinforcing the friction interpretation.
- **Result 3: Scope matters.** Groups with more workers facing the same tax incentives have larger observed elasticities.
- **Result 4: Firms and unions matter.** Occupation-level wage modes and teacher wage distributions show that job offers are shaped around common tax incentives.
- **Result 5: Individual and aggregate bunching are related.** Occupations with more individual bunching at the pension kink show more aggregate bunching at the statutory kink.
- **Result 6: Self-employed behavior is a placebo.** The self-employed bunch at both small and large kinks and do not exhibit the same scope patterns as wage earners.

9 Short summary

- The paper studies why micro estimates of labor supply elasticities are often small while macro evidence suggests larger responses.
- The mechanism is a combination of worker adjustment costs and firm/union hours constraints.
- Large tax kinks can overcome search costs; small tax reforms often cannot.
- Firm and union responses can create aggregate bunching when many workers share a tax incentive.
- Danish administrative records allow the authors to test size, scope, occupation, and self-employment predictions.

- Wage earners bunch at the large top tax kink but not much at the small middle kink.
- Small tax-reform estimates are near zero.
- Teachers and occupation-level wage modes reveal aggregate bunching around common tax thresholds.
- The pension kink separates individual adjustment from aggregate firm response.
- Self-employed individuals respond strongly at both small and large kinks, supporting the role of wage-earner frictions.
- The paper cautions that small quasi-experiments may understate steady-state effects of broad policy reforms.

10 Conclusion

10.1 Core conclusion

Chetty, Friedman, Olsen, and Pistaferri show that observed labor-supply elasticities are not invariant parameters. They depend on the policy environment, the magnitude of the incentive, the number of workers affected, and the extent to which firms and institutions adjust job offers.

10.2 Economic intuition

If workers must pay search costs to change jobs or hours, small incentives are not enough to trigger adjustment. Large tax changes or broad reforms can alter both individual search behavior and the equilibrium menu of jobs. As a result, micro estimates from small tax changes can be much smaller than elasticities relevant for broad, long-run policy comparisons.

10.3 Contribution to public finance

The paper provides a way to interpret the common lack of bunching at many tax kinks without concluding that labor supply is completely inelastic. It shows that the absence of bunching can reflect frictions rather than zero preferences for adjustment.

10.4 Contribution to labor economics

The paper emphasizes that hours and compensation packages are not chosen unilaterally by workers. Firms and unions create the feasible set, and their responses can shape the labor-supply effects of taxes.

10.5 Policy implication

The efficiency cost of a tax reform may differ depending on whether the reform affects a small group or the whole economy. Policies that apply broadly can change wage-setting institutions and job menus, producing larger long-run responses than small-group quasi-experiments suggest.

10.6 Limitation

The paper does not estimate the structural macro elasticity directly. It also studies taxable income and wage earnings rather than hours. Nevertheless, the evidence strongly indicates that adjustment costs and firm responses attenuate standard micro estimates.

10.7 Takeaway

The central lesson is that labor supply responses are mediated by frictions and institutions. Small tax changes reveal short-run, local responses. Large and broad policy changes may reveal a different, more macro-relevant elasticity.